

AVAGENT: Bridging Asynchronous Communication Through AI-Powered Virtual Avatars

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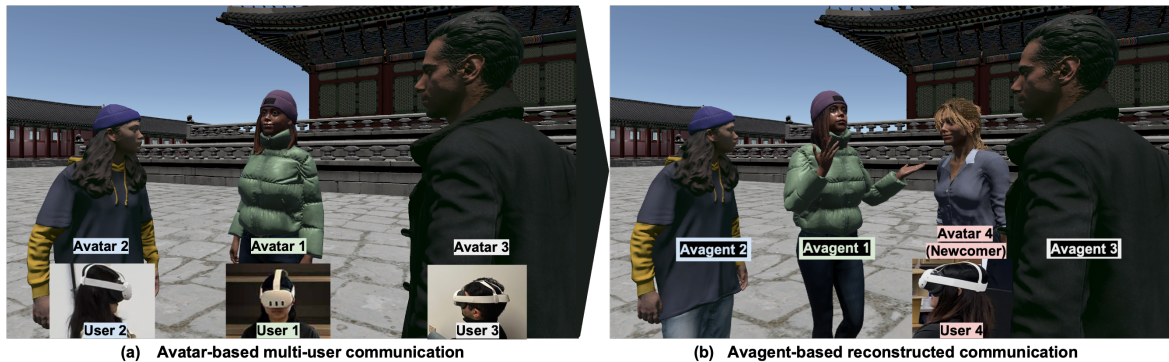


Figure 1: Example of Avagent in a meeting: (a) avatar-based multi-user interaction during the initial communication, and (b) avagent-based reconstructed interaction, where an agent enhances the experience for a newcomer based on the initial interaction.

ABSTRACT

Asynchronous communication is essential in diverse contexts such as remote meetings, education, and collaboration. However, traditional replay-based systems lack interactivity and often fail to preserve critical contextual cues. This paper introduces *Avagent*, a novel framework for asynchronous communication in Virtual/Augmented Reality (VR/AR) environments that leverages agentized avatars to bridge temporal and contextual gaps between participants. Unlike typical agent-mediated systems, the framework utilizes users' data to create Avagents that reflect their actual intent, emotional states, and behaviors. By accurately capturing and replicating both verbal and nonverbal behaviors, Avagents enable more interactive and context-rich communication. This approach facilitates dynamic interactions, enhances social presence, and ensures continuity across timelines. Avagents empower new users to engage interactively with prior discussions, fostering deeper understanding and seamless collaboration with previous users. Envisioning the benefits of Avagents as an engaging and context-rich solution for asynchronous communication, this paper outlines interaction scenarios, a work-in-progress prototype, and associated challenges.

Index Terms: Virtual reality, Asynchronous Communication, Virtual Avatar, Virtual Agent.

1 INTRODUCTION

Immersive technologies, including Virtual and Augmented Reality (VR/AR), have transformed remote communication by overcoming the constraints of physical space. Platforms such as Meta Horizon

Workrooms¹ and Spatial² enable users to collaborate in immersive virtual environments. By utilizing embodied virtual avatars that convey both verbal and nonverbal cues, these platforms support meetings and teamwork in ways that were previously unimaginable. However, these platforms lack the capabilities to adequately address *temporal gaps* when participants cannot attend meetings simultaneously. This limitation is particularly problematic in contexts, such as in global teams, remote education, and international research collaborations [1, 13]. In today's interconnected world, the need for systems that can synchronize experiences and maintain shared understanding across different timelines has become increasingly evident.

To address these challenges, this position paper introduces the concept of *Avagent*—a novel framework that leverages *agentized* avatars powered by Artificial Intelligence (AI) to bridge temporal and contextual gaps in asynchronous communication. Avagents are intelligent, interactive virtual agents that go beyond merely presenting previously recorded meeting or experience data (Figure 1). They interpret and reconstruct verbal and nonverbal data/information, enabling subsequent participants to interactively engage with the content and even generate new insights. By capturing participants' verbal behaviors (e.g., speech content) and nonverbal behaviors (e.g., gestures and facial expressions), the proposed system fosters interactive, context-rich communication between initial and later participants.

This Avagent framework aims to enhance the synchronization of experiences, foster dynamic interactions, and create scalable solutions that enable meaningful communication across temporal barriers while being guided by the following research questions (RQs):

RQ1. How can the *agentization* of avatars' verbal and nonverbal data in VR/AR environments enhance contextual understanding in asynchronous communication?

RQ2. How does interaction with agentized avatars impact user

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¹Meta Horizon Workrooms: <https://forwork.meta.com/ca/horizon-workrooms/> (Accessed 2024-12-14)

²Spatial: <https://www.spatial.io/> (Accessed 2024-12-14)

engagement and participation in asynchronous communication?

RQ3. How can agentized avatars improve user interaction and information transfer efficiency during synchronization in asynchronous communication?

By addressing these RQs, the framework lays the groundwork for a transformative approach to asynchronous communication in VR/AR environments. This paper outlines the design framework of Avagents and explores their potential applications and implications across various domains.

2 RELATED WORK

2.1 Asynchronous Communication

Research on asynchronous communication has explored various approaches to improve user experience in scenarios where participants interact across different timeframes [8]. These studies focus on enhancing the effectiveness of information delivery, contextual understanding, and user engagement in asynchronous environments [2, 15]. Based on these features, there has been active research exploring its applicability in various industries [6, 7]. However, existing systems often rely on static recordings, limiting interactivity and dynamic communication among participants.

Current asynchronous communication systems rely predominantly on static recordings, such as replay-based systems or simple chatbot functionalities [5, 10, 12]. While these systems offer a basic means to review past discussions, they face critical limitations. Static recordings fail to convey the full context of interactions, such as subtle gestures, emotional tones, or implicit meanings, leaving later participants disconnected from the original discussion's nuances. Furthermore, these systems lack interactivity, preventing subsequent participants from asking follow-up questions, clarifying points of confusion, or exploring additional details. This lack of engagement reduces collaboration effectiveness, particularly in VR/AR environments, where nonverbal cues play a crucial role in communication and understanding [17].

2.2 Agent Development Using Avatar Data

There has been extensive research on creating automated agents based on user data. In particular, studies have actively explored developing agents that utilize users' behavioral information, such as body movements, captured through IoT devices and cameras, for various applications [11, 14, 9].

Furthermore, there has been growing interest in leveraging avatar data to develop intelligent agents that facilitate user interaction. These studies utilize verbal and nonverbal data, such as speech, gestures, and expressions, captured from avatars to create agents capable of responding to user queries and providing contextual insights [16]. Recently, advancements in computing capabilities, particularly those based on Large Language Models (LLMs) capable of processing multimodal data, have further accelerated research in this area [18]. Such approaches aim to bridge the gap between static data representation and interactive, adaptive systems, enabling richer communication experiences in virtual environments.

3 AVAGENT: ASYNCHRONOUS COMMUNICATION SYSTEM

3.1 Overview

The proposed framework addresses the critical challenges of asynchronous communication in VR/AR environments by leveraging agentized avatars (Avagents) to bridge temporal and contextual gaps between participants (Figure 2). This system enables dynamic and interactive communication between users who participate in the 'Initial Communication (avatar-based multi-user interaction)' and those who join asynchronously in the 'Agent-Enhanced Communication (reconstructed 'Initial Communication' with Avagents)'.

The framework consists of five distinct phases, each designed to ensure continuity, engagement, and contextual understanding across timelines:

- **Recording Communication** records participants' verbal and nonverbal behaviors during the 'Initial Communication'.
- **Generating Avagent Interactions** transforms the recorded data into interactive agentized avatars (Avagents).
- **Interacting with a Newcomer** replays the recorded 'Initial Communication' while allowing a newcomer to interact with the Avagents dynamically during the 'Agent-Enhanced Communication'.
- **Syncing with Users from the 'Initial Communication'** involves Avagents summarizing and explaining their interaction with the newcomer during the 'Agent-Enhanced Communication' to the participants from the 'Initial Communication'.
- **Achieving Full Synchronization** integrates all participants by facilitating direct communication between them—i.e., both the participants from the 'Initial Communication' and the newcomer introduced during the 'Agent-Enhanced Communication'.

This approach enables communication sync-up across all participants, including the emotional context, regardless of whether they attended the 'Initial Communication' or the 'Agent-Enhanced Communication'. Furthermore, this framework supports scalability, as new participants can seamlessly understand the content and context of the discussion.

3.2 Framework Design

Phase 1: Recording Communication This phase involves collecting interaction data from participants during the 'Initial Communication'. Data includes verbal components (speech content) and nonverbal components (body movements and facial expressions). This data serves two purposes: reconstructing the 'Initial Communication' for later review and processing the data to create Avagents. This phase ensures that the raw interaction data is both preserved for future use and prepared for transformation into dynamic, context-aware agents.

Phase 2: Generating Avagent Interactions The second phase focuses on transforming the recorded data into interactive Avagents capable of engaging with subsequent participants during the 'Agent-Enhanced Communication'. These Avagents are designed to replicate the behaviors of the original participants and autonomously facilitate meaningful interactions. Key processes in this phase include:

- **Verbal Interaction Generation:** Speech data is synthesized into natural language responses, allowing Avagents to answer questions, provide clarifications, and expand on discussion points.
- **Nonverbal Interaction Generation:** Gestures, facial expressions, and other nonverbal behaviors are recreated to mirror the emotional and contextual cues of the original participants.

Phase 3: Interacting with a Newcomer The third phase focuses on enabling a newcomer (i.e., a participant who was not part of the 'Initial Communication') to gain a deep understanding of the original discussion through dynamic interaction with Avagents. In this phase, the newcomer reviews the reconstructed meeting by engaging directly with the Avagents designed to replicate the verbal and nonverbal behaviors of the original participants from the 'Initial Communication'. These Avagents provide the newcomer with a context-rich experience, preserving the emotional and informational nuances of the original meeting. The newcomer can interact

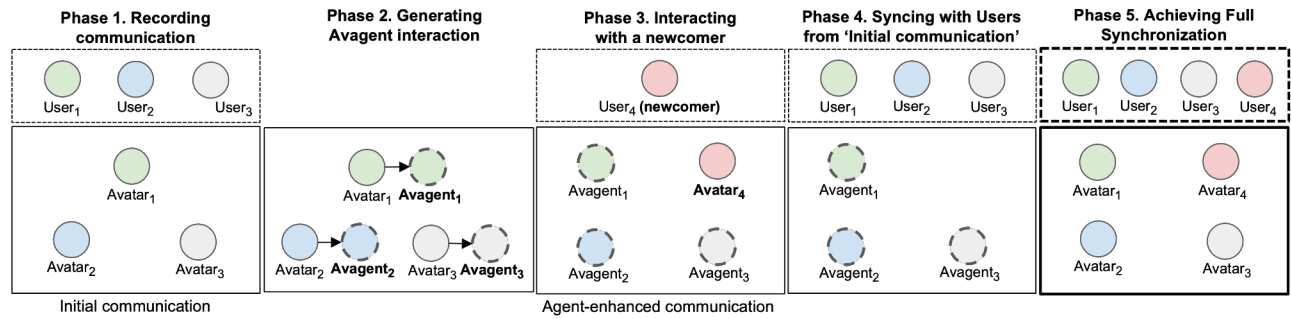


Figure 2: Framework Overview: Phase 1. Recording Communication, Phase 2. Generating Avagent Interactions, Phase 3. Interacting with a Newcomer, Phase 4. Syncing with Users from 'Initial Communication', and Phase 5. Achieving Full Synchronization.

with the Avagents by asking follow-up questions, seeking clarifications, and exploring discussion topics in greater depth. This dynamic interaction allows the newcomer to actively participate in the communication process rather than passively consuming recorded content. By fostering a sense of social presence and inclusion, Avagents create an immersive experience that bridges the temporal gap, enabling asynchronous users to fully engage with the meeting content, and contribute effectively to the ongoing collaboration.

Phase 4: Syncing with Users from 'Initial Communication' This phase synchronizes the new interactions from the 'Agent-Enhanced Communication' with the original participants of the 'Initial Communication'. This ensures that the contributions of subsequent users (i.e., the newcomer) are seamlessly integrated into the overall communication flow. During this phase, the original participants can review the interactions that occurred between their Avagents and the newcomer. Additionally, updates, such as new questions, topics, or insights, are conveyed to them through their own Avagents, creating a cohesive and unified communication experience.

Phase 5: Achieving Full Synchronization In this phase, users from each communication session come together to interact, either in person or online via avatars. Despite their interactions occurring at different times, both parties are already aligned with the context and intent of the communication. Through prior synchronization facilitated by Avagents, the framework ensures that all users, regardless of their timeline, can review the interactions and reach a consensus on the discussion. This process preserves the emotional and informational continuity of the communication, fostering seamless collaboration and engagement.

3.3 Potential Use Cases

To illustrate the functionality and potential of the proposed framework, the following mock-up scenarios demonstrate how Avagents can be applied in real-world settings:

Corporate Collaboration A global company conducts a multi-user meeting where a team leader from another time zone is unable to attend. Later, the leader uses the Avagent system to review the meeting. Through the Avagent, they engage with the discussion by asking follow-up questions and exploring critical topics. The responses generated by the Avagent retain the context and nuances of the original conversation, enabling the leader to contribute effectively to the project.

Education In a virtual classroom, a student misses a live lecture due to a scheduling conflict. Using the Avagent system, the student can review the lecture content interactively, asking the Avagent for clarifications or additional examples. The system replicates the lecturer's verbal and nonverbal cues, creating a more personalized and immersive learning experience.

Healthcare Training In a healthcare training session, a medical trainee who missed a live VR-based simulation interacts with an Avagent representing the instructor. The Avagent guides the trainee through key scenarios, such as responding to emergencies or practicing diagnostic techniques, while answering questions and offering tailored feedback. By replicating the instructor's verbal and nonverbal cues, the system provides an engaging and realistic training experience, ensuring the trainee gains essential skills and knowledge without requiring the instructor's real-time presence.

Customer Service A company leverages the Avagent framework to offer advanced customer support in a VR-based environment. Customers interact with Avagents representing support staff, receiving tailored, context-aware assistance with emotional engagement. The Avagent system replicates the communication style of the support staff, providing a personalized and engaging experience while ensuring consistent and scalable service, even when real staff members are unavailable.

4 PROTOTYPE DEVELOPMENT

The initial prototype of the Avagent framework is in the middle of development using advanced tools to create realistic, interactive virtual human characters and support dynamic asynchronous communication in VR/AR environments (Figure 3). The prototype leverages Avaturn³ for avatar creation, Unity for system development, and the Meta Quest 3 headset for deployment. The development consists of two primary components:

Recording and Replaying Avatar-Based Interaction

The first component focuses on capturing and replaying the avatar owner's verbal and nonverbal behaviors during a VR interaction. Verbal behaviors, including speech, are transcribed into text with speech-to-text (STT) technology (OpenAI's Whisper⁴), while nonverbal behaviors, such as body gestures, are recorded using the Meta Quest Controllers and Inside-Out Body Tracking (IOBT). These recorded interactions are stored and can be replayed, providing the foundation for generating interactive Avagents.

Generating Avagent Verbal and Nonverbal Behaviors

The second component transforms the recorded data into dynamic Avagent behaviors, allowing users to interact with reconstructed content. The implementation includes:

- **Verbal Interaction Generation:** To create new verbal interactions, the recorded speech content is used as input for a

³Avaturn: <https://avaturn.me/> (Accessed 2024-12-22).

⁴Whisper: <https://openai.com/index/whisper/> (Accessed 2024-12-22).

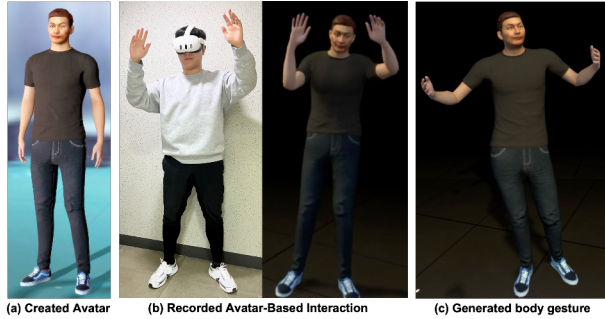


Figure 3: Prototype development (Example): (a) created avatar from Avaturn, (b) Recorded avatar-based interaction with Meta Quest, and (c) Generated body gesture with ZeroEGGS.

Large Language Model (LLM, OpenAI's GPT-4o⁵), to generate dialogue for the Avagent. The generated content is synthesized into audio using text-to-speech (TTS) technology. A custom voice clone, generated using ElevenLabs' voice cloning technology⁶, ensures that the Avagent's verbal feedback accurately reflects the original speaker's voice.

- **Nonverbal Interaction Generation:** Gestures and other non-verbal cues are generated in sync with the audio using ZeroEGGS⁷, a zero-shot example-based gesture generation tool. This ensures that the Avagent's physical expressions align with the intended emotional and contextual nuances.

By integrating these components, the prototype demonstrates the potential of Avagents to provide dynamic, context-rich interactions that bridge temporal gaps in asynchronous communication. Future iterations will focus on refining these functionalities, enhancing user interfaces, and addressing identified challenges to create a robust, user-centric communication system.

5 DISCUSSION

The proposed Avagent framework uses AI-powered avatars to bridge temporal gaps in asynchronous communication, transforming static systems into dynamic, engaging, and context-rich environments. This section discusses the implications of this novel approach, focusing on its potential to enhance user interaction and collaboration across various domains, while also addressing the challenges and limitations.

5.1 Implications

In addition to facilitating informational continuity, Avagents play a crucial role in enhancing social presence and emotional connection. In traditional asynchronous systems, users often experience a sense of isolation when consuming passive recordings, such as online learning [4]. In contrast, interacting with Avagents creates an immersive experience that mirrors the emotional engagement of real-time communication. This enhancement of social presence is pivotal in VR/AR environments, as it sustains user attention, motivation, and participation, even in asynchronous settings [3].

The proposed framework is not limited to a single domain; its scalability allows for broader applications across various fields. Beyond its immediate relevance to remote meetings and global collaboration, this system could be adapted for education, where

⁵GPT-4o: <https://openai.com/index/hello-gpt-4o/> (Accessed 2024-12-22).

⁶ElevenLabs: <https://elevenlabs.io/> (Accessed 2024-12-22).

⁷ZeroEGGS: <https://github.com/ubisoft/ubisoft-laforge-ZeroEGGS?tab=readme-ov-file> (Accessed 2024-12-22).

it supports interactive learning between students and instructors; in healthcare, where it enables non-contact counseling or medical training; and in customer service, where it offers personalized and context-aware interactions, as described in Section 3.3. The flexibility and scalability of this approach highlight its potential to transform asynchronous communication across diverse sectors.

By addressing the limitations of existing systems and introducing an innovative, interactive, and scalable framework, this paper seeks to transform asynchronous communication in VR/AR environments.

5.2 Challenges and Limitations

The proposed framework for asynchronous communication using agentized avatars (Avagents) introduces several technical and operational challenges that must be addressed to ensure its effectiveness and scalability. One major challenge lies in the accurate real-time processing and replication of verbal and nonverbal interaction data. Errors in speech synthesis, gesture recreation, or data processing could compromise the quality of Avagents, leading to ineffective or misleading interactions. Additionally, balancing the personalization of Avagents with the scalability required for multi-user and cross-cultural applications presents a significant hurdle. The computational demands of real-time interaction generation and synchronization further complicate implementation, especially in resource-constrained environments such as mobile VR/AR devices.

Another concern is the ethical and security implications of handling user data. Capturing sensitive data such as voice, gestures, and facial expressions raises privacy issues, requiring robust encryption and secure storage mechanisms to prevent unauthorized access or misuse. Furthermore, Avagents should be safeguarded against tampering or impersonation to maintain trustworthiness and reliability. Ethical dilemmas, such as the potential misuse of avatars to manipulate emotional responses or replicate behaviors without consent, underscore the need for clear guidelines and regulations. Addressing these challenges will require a multi-disciplinary approach, combining advances in AI, data security, and user-centric design to ensure the framework's adoption and long-term viability.

6 CONCLUSION

The proposed framework transforms asynchronous communication in VR/AR environments by shifting from static, replay-based systems to dynamic, interactive experiences. Leveraging agentized avatars (Avagents), the system addresses critical challenges such as context preservation, lack of interactivity, and reduced engagement. The approach involving five phases—Recording Communication, Generating Avagent Interactions, Interacting with a Newcomer, Syncing with Users from 'Initial Communication', and Achieving Full Synchronization described in Section 3.2—ensures seamless synchronization of experiences and fosters meaningful collaboration across temporal barriers.

This framework not only enhances user engagement and social presence but also demonstrates broad applicability across domains, including education, healthcare, and corporate collaboration. Despite challenges in data processing, scalability, and privacy, this innovative approach highlights the potential of AI-powered agents to reshape asynchronous communication. Future research will focus on completing the development of Avagents by extending the current prototype. Efforts will prioritize refining functionalities for temporal synchronization and user interfaces, addressing ethical concerns, and expanding applications to create a robust, user-centric communication ecosystem. By bridging the temporal gaps in VR/AR environments, the framework sets a new standard for immersive, asynchronous collaboration.

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